

Homework 8

Due: Friday, 26th July 2024 at 1:00 PM

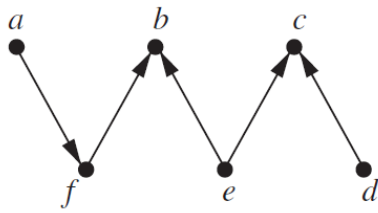
Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question (except the extra credit question) is worth 10 points.

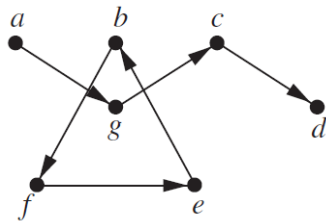
Question 1.

Determine whether the following graph is strongly connected and if not, whether it is weakly connected.



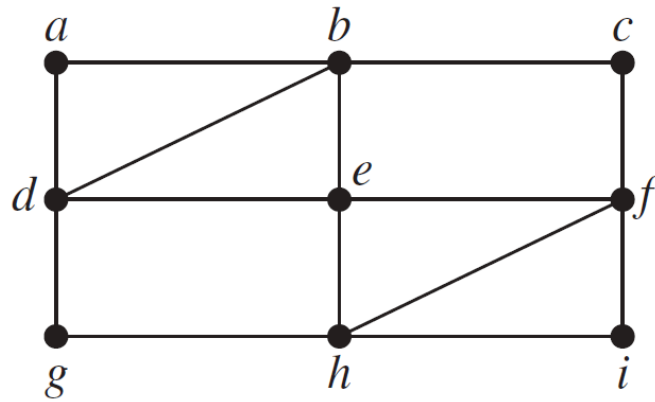
Question 2.

Determine whether the following graph is strongly connected and if not, whether it is weakly connected.

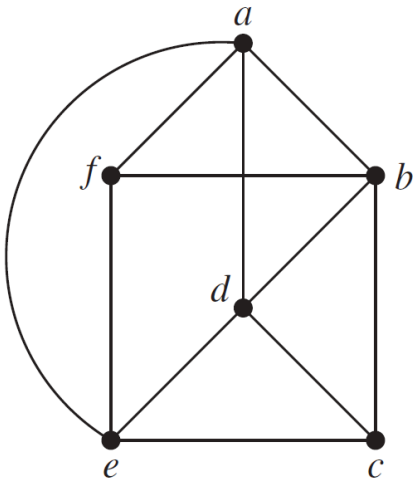


In Exercises 3–6 determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists.

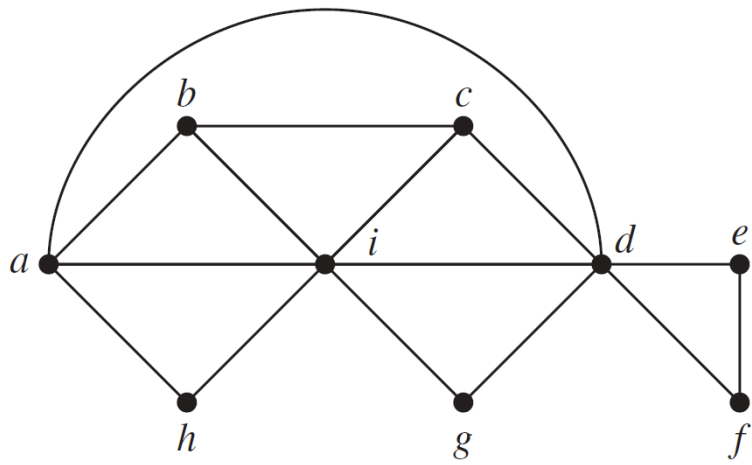
Question 3.



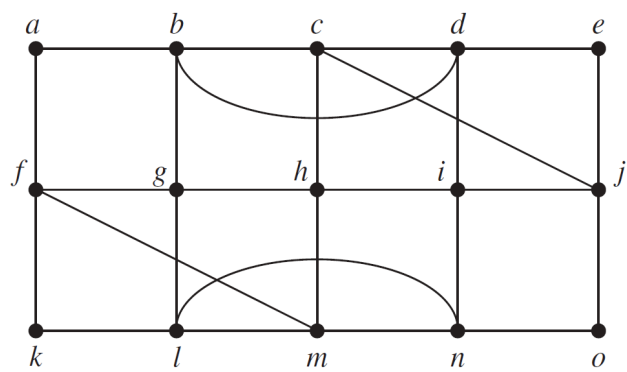
Question 4.



Question 5.

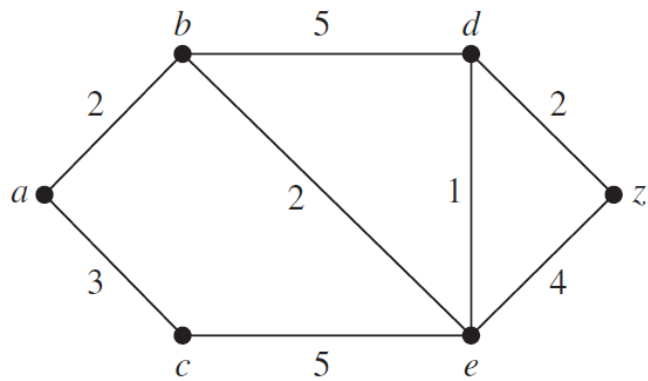


Question 6.



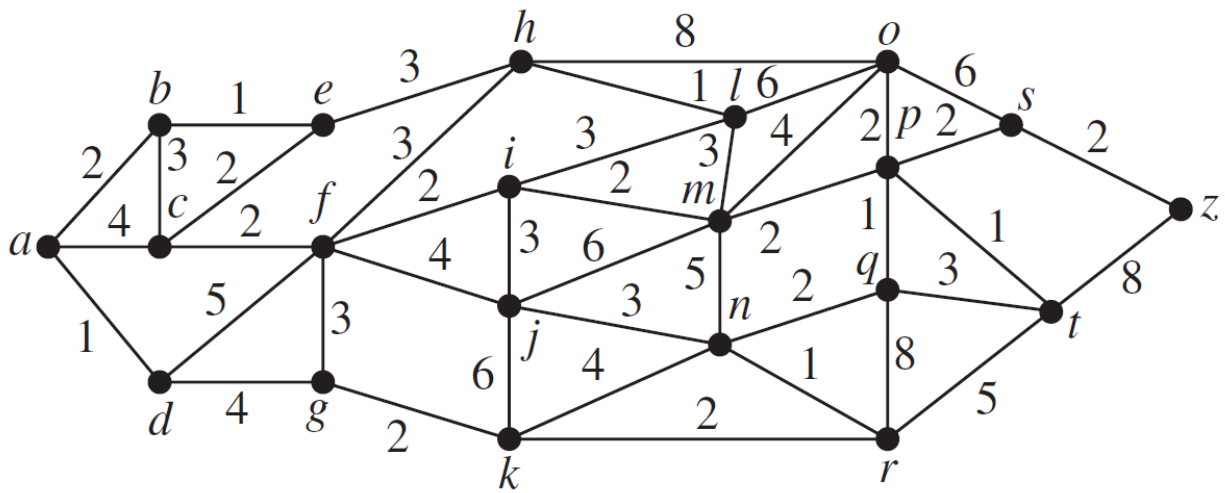
Question 7.

Find the length of a shortest path between a and z in the given weighted graph.



Question 8.

Find the length of a shortest path between a and z in the given weighted graph.



Question 9.

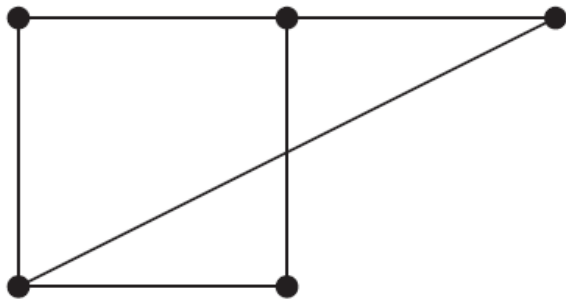
Extend Dijkstra's algorithm for finding the length of a shortest path between two vertices in a weighted simple connected graph so that a shortest path between these vertices is constructed.

Question 10.

Is a shortest path between two vertices in a weighted graph unique if the weights of edges are distinct?

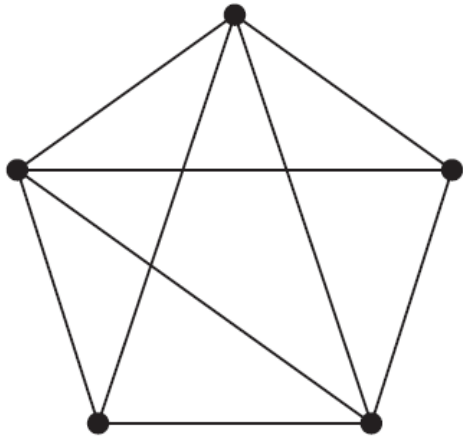
Question 11.

Draw the given planar graph without any crossings.



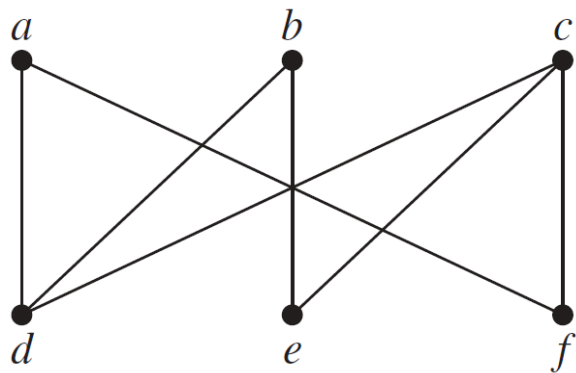
Question 12.

Draw the given planar graph without any crossings.



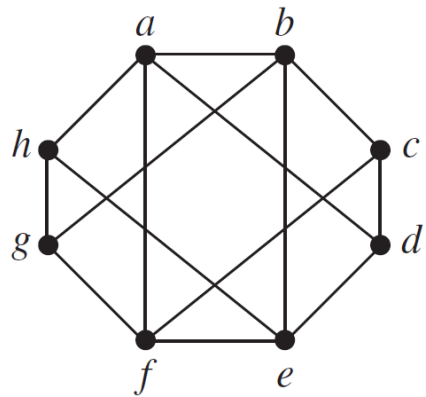
Question 13.

Determine whether the given graph is planar. If so, draw it so that no edges cross.



Question 14.

Determine whether the given graph is planar. If so, draw it so that no edges cross.



Question 15.

Suppose that a connected planar graph has eight vertices, each of degree three. Into how many regions is the plane divided by a planar representation of this graph?

Extra credit

Show that the problem of finding a circuit of minimum total weight that visits every vertex of a weighted graph at least once can be reduced to the problem of finding a circuit of minimum total weight that visits each vertex of a weighted graph exactly once. Do so by constructing a new weighted graph with the same vertices and edges as the original graph but whose weight of the edge connecting the vertices u and v is equal to the minimum total weight of a path from u to v in the original graph.

Notice that, if you answer the extra credit question correctly, the grade of the extra credit will not show up on Gradescope. But we will add that grade in D2L at the end of the semester.